Sri Lanka Institute of Information Technology

Assignment 2

IT3021 - Data Warehousing and Business Intelligence

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1. Overview of the Data Source

The data source used for Assignment 2 is the **CarSalesDW** data warehouse designed and implemented in Assignment 1. It is built on SQL Server (Express Edition) and is based on the AutoBid Analytics used vehicle auction scenario. The warehouse stores wholesale vehicle auction transaction data comprising over 600,000 records and supports multi-dimensional analytical processing.

The primary raw data was sourced from a real-world Kaggle dataset (Used Vehicle Sales Data) and subsequently extended with synthetically generated records to cover a complete year of transactions.

1.1. Recommended Scenario: Used Vehicle Auction & Sales Processing System.

Scenario Title: Used Vehicle Auction Lifecycle & Market Pricing Intelligence - AutoBid Analytics

AutoBid Analytics is a regional used vehicle auction house that sources pre-owned vehicles from franchise dealerships, financial institutions (repossessions), and fleet companies. Each vehicle is registered in the company's internal Dealer Management System (DMS), undergoes inspection, is listed for auction, sold, and subsequently transferred to the buyer. In parallel, an external Market Pricing API provides daily Manheim Market Report (MMR) reference prices in CSV format.

The primary data source for this study is derived from a real-world vehicle sales dataset obtained from **Kaggle.** The original dataset contains over 558,000 records with multiple attributes, including vehicle specifications, seller details, market valuation (MMR), actual selling price, and transaction dates.

To meet the analytical requirements of this project, the dataset was extended and refined into **car_prices_extended.csv**. This enhanced dataset contains over 600,000 records and ensures a complete year of transactional data. The extension was performed using synthetic data generation techniques while preserving realistic relationships and distributions within the data.

The final dataset represents wholesale vehicle auction transactions and includes key attributes such as vehicle characteristics (make, model, trim, body type, transmission), condition indicators (odometer readings and condition scores), seller information, geographic location (state), estimated market value (MMR), and final selling price.

Original dataset link - https://www.kaggle.com/datasets/syedanwarafridi/vehicle-sales-data

1.2 ER-Diagram for AutoBid Analytics Vehicle Auction System

This ER diagram represents the conceptual data model of the AutoBid Analytics vehicle auction system. It illustrates the key entities, attributes, and relationships involved in managing vehicle auctions, inspections, sellers, locations, and sales transactions.

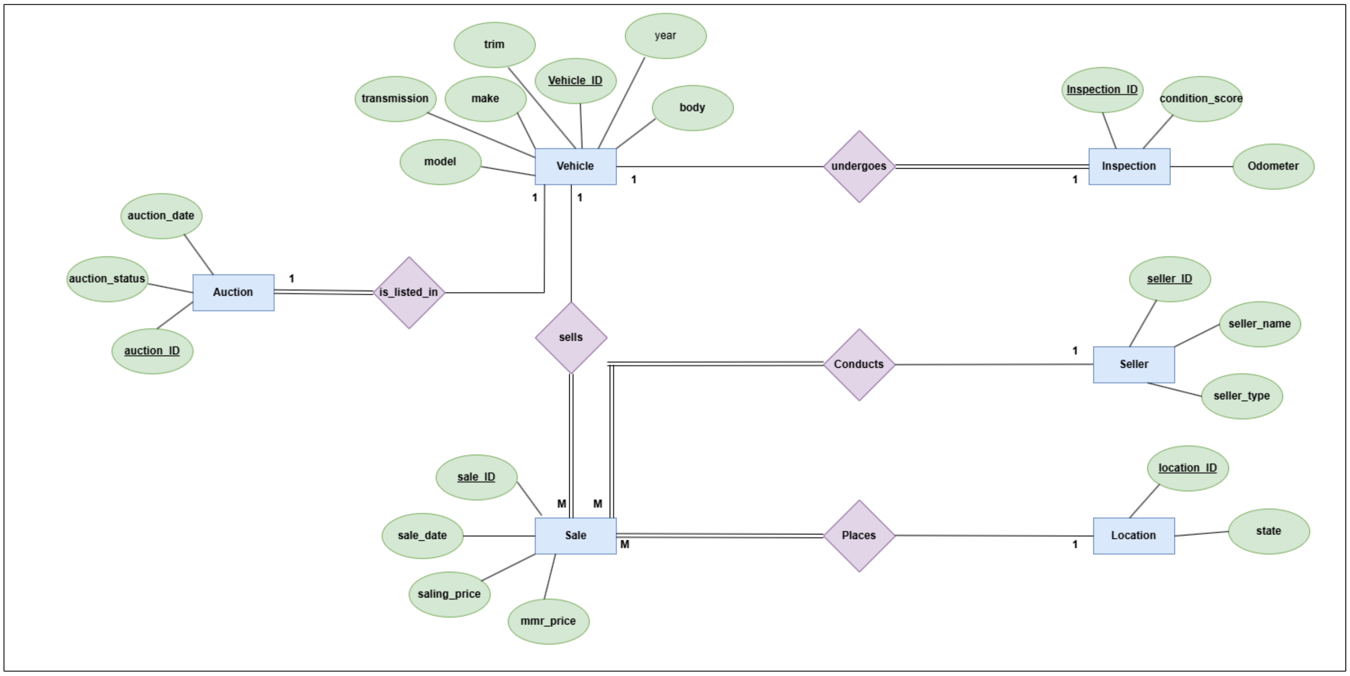


Figure 1 – ER Diagram for AutoBid Analytics Vehicle Auction System

Assumptions

1. Each vehicle is assumed to undergo at most one inspection prior to sale. Therefore, a **one-to-one relationship** between Vehicle and Inspection is considered, even though real-world scenarios may involve multiple inspections.
2. The dataset contains only the final selling price of vehicles. Hence, it is assumed that **each sale represents the outcome of an auction**, and intermediate bidding details are not modeled.
3. Each sale is assumed to be conducted by exactly one seller. However, a seller may conduct multiple sales over time, resulting in a **one-to-many relationship between Seller and Sale**.
4. Each sale is assumed to occur at a single location (state). A location can have multiple sales, but each sale is linked to only one location.
5. A vehicle may or may not be sold. However, each sale must be associated with exactly one vehicle. Therefore, the relationship between Vehicle and Sale is modeled as **one-to-many with total participation from the Sale side**.
6. It is assumed that the dataset contains sufficient and valid data for vehicle attributes, seller details, pricing, and dates. Missing or inconsistent values are handled during the data preprocessing stage.
7. Since the original dataset did not cover a full year, additional records were generated synthetically. It is assumed that the generated data maintains **realistic distributions and relationships** consistent with the original dataset.

8. Surrogate keys (e.g., vehicle_ID, seller_ID) are introduced to uniquely identify records and support efficient database design, as the original dataset does not provide consistent primary keys.

1.3 Data Warehouse Design

A dimensional data warehouse model was designed using a star schema to support analytical processing. The design includes a central fact table representing vehicle sales transactions, surrounded by four-dimension tables such as Vehicle, Seller, Location, and Date. This design was chosen due to its simplicity, query performance efficiency, and direct suitability for OLAP operations.

The model was structured to ensure:

1) Clear definition of grain (one record per vehicle sale)
2) Proper use of surrogate keys
3) Separation of facts and dimensions
4) Support for efficient querying and reporting

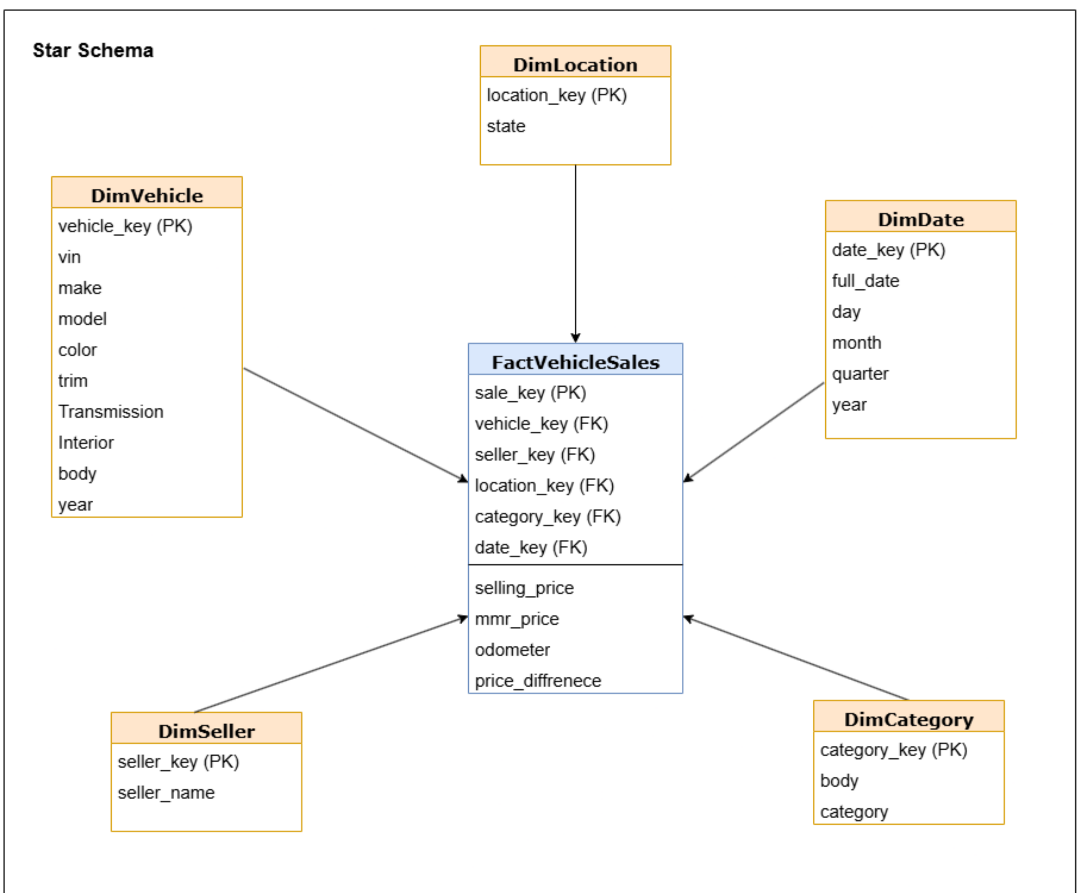


Figure 2 - Star Schema Design of Data Warehouse

The architecture includes five main layers:

- Data Sources – sale_data.xlsx (Excel), vehicle_data.csv, vehicle_category.csv
- Staging Area – Temporary storage for initial data validation and cleansing
- ETL Process – Implemented using SQL Server Integration Services (SSIS)
- Data Warehouse (CarSalesDW) – Star schema with fact and dimension tables
- BI / Reporting Layer – Power BI dashboards and SSAS OLAP cubes

1.4 Data Warehouse Tables

Table Name	Type	Primary Key	Key Attributes
Fact_VehicleSales	Fact Table	sale_key	vehicle_key, seller_key, date_key, location_key, category_key
Dim_Vehicle	Dimension	vehicle_key	make, model, trim, body, transmission, year, color, interior
Dim_Seller	Dimension	seller_key	seller_name, seller_type, effective_date, end_date, is_current
Dim_Date	Dimension	date_key	full_date, year, month, day
Dim_Location	Dimension	location_key	state
Dim_Category	Dimension	category_key	body_type, category

1.5 Fact Table Fact_VehicleSales

The Fact_VehicleSales table is the central table of the data warehouse. It stores quantitative measures related to each vehicle sale transaction. The grain of the fact table is defined as one record per completed vehicle sale transaction.

The table was later extended into an Accumulating Snapshot Fact Table by adding accm_txn_create_time, accm_txn_complete_time, and txn_process_time_hours columns to track the full lifecycle of each transaction.

Measure	Data Type	Description
selling_price	DECIMAL (10,2)	Actual sale price of the vehicle at auction
mmr	DECIMAL (10,2)	Manheim Market Report reference price
odometer	INT	Vehicle mileage at time of sale
condition_score	INT	Numeric condition rating of the vehicle prior to sale
Price_difference	Computed column	Calculated as selling_price minus mmr; measures pricing deviation

1.6 Dimension Tables

1.Dim_Vehicle

Stores detailed attributes of each vehicle involved in a sale. Supports analysis by brand, model, trim level, body type, and year.

- Attributes: vehicle_key (PK), make, model, trim, body, transmission, year, color, interior

2.Dim_Seller (SCD Type 2)

Stores seller information. Implemented as a Slowly Changing Dimension Type 2 to track historical changes in seller attributes (e.g., seller classification changes over time).

- Attributes: seller_key (PK), seller_name, seller_type, effective_date, end_date, is_current

3.Dim_Date

Supports time-based analysis and drill-down operations across the date hierarchy. Date attributes were derived from the sale date during ETL transformation.

- Attributes: date_key (PK), full_date, year, month, day

4.Dim_Location

Stores geographical information to support regional sales analysis.

- Attributes: location_key (PK), state

5.Dim_Category

Categories vehicles by body type (e.g., SUV -> Utility, Sedan -> Passenger Car). Enables grouping and analysis of vehicle types.

- Attributes: category_key (PK), body_type, category

1.7 Hierarchies

The data warehouse incorporates the following hierarchies to support drill-down and roll-up OLAP operations

Hierarchy	Dimension	Levels
Date Hierarchy	Dim_Date	Year → Month → Day
Vehicle Hierarchy	Dim_Vehicle	Make → Model → Trim

1.8 Supported Aggregations

The data warehouse supports the following aggregate measures, enabling comprehensive business analysis:

- Total sales amount per seller, location, vehicle category, or time period
- Average selling price compared against MMR (market value)
- Total number of vehicles sold across dimensions
- Price difference analysis (actual selling price vs. MMR benchmark)
- Average vehicle condition score by make/model/year
- Sales transaction lifecycle duration (txn_process_time_hours)

1.9 Database details

The table below summaries the key technical details of the CarSalesDW data warehouse

Property	Details
Database Name	CarSalesDW
Database System	Microsoft SQL Server (Express Edition)
Schema Type	Star Schema
Fact Table	Fact_VehicleSales
Number of Dimensions	5 (Vehicle, Seller, Date, Location, Category)
Record Volume	600,000+ vehicle sale transactions
ETL Tool	SQL Server Integration Services (SSIS)
Data Sources	sale_data.xlsx, vehicle_data.csv, vehicle_category.csv
SCD Implementation	Dim_Seller – Type 2 (effective_date, end_date, is_current)
Fact Table Type	Transactional + Accumulating Snapshot

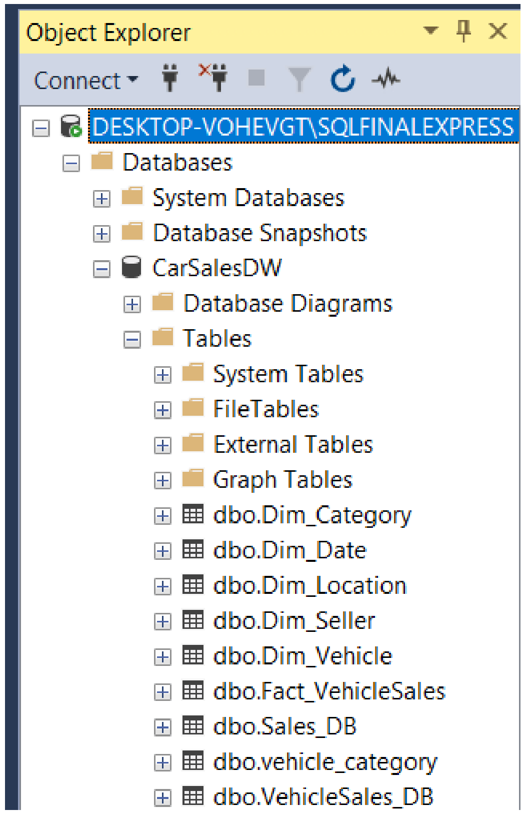


Figure 3 - CarSalesDW SSMS Object Explorer

2. SSAS Cube Implementation

An SSAS (SQL Server Analysis Services) Multidimensional Cube was created using Visual Studio 2022 with the Analysis Services extension. The data warehouse CarSalesDW, hosted on the DESKTOP-VOHEVGT\SQLFINALEXPRESS SQL Server instance, was used as the data source. The cube was deployed to the DESKTOP-VOHEVGT\SQLDEV Analysis Services instance.

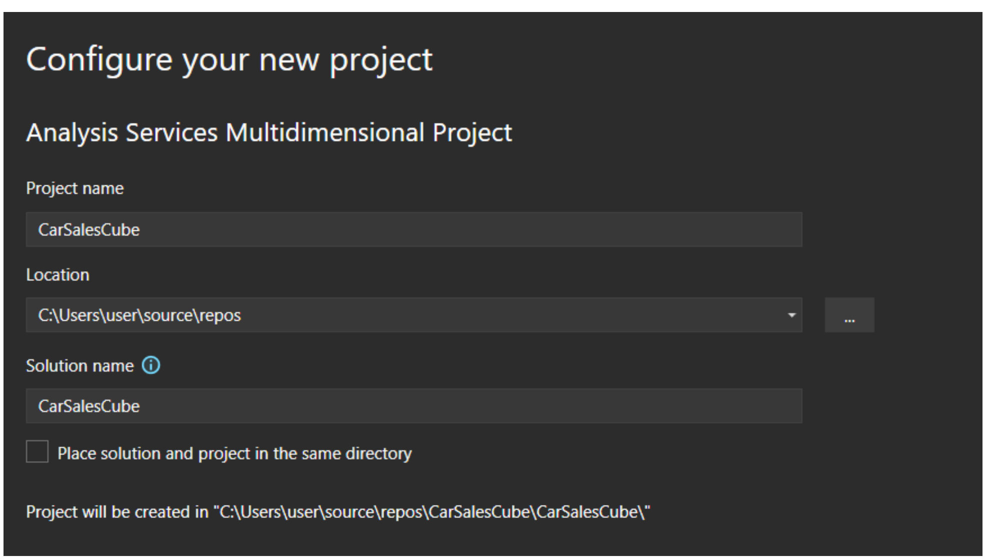


Figure 4 - Create new Analysis Services Multidimensional Project

Visual Studio 2022 was opened, and a new project was created by selecting 'Analysis Services Multidimensional Project' from the project templates. The project was named CarSalesCube.

2.1 Create Data Source

A new Data Source was created by right clicking the 'Data Sources' folder in Solution Explorer and selecting 'New Data Source'. The Data Source Wizard was used to configure the connection.

- Provider: Native OLE DB\Microsoft OLE DB Driver for SQL Server
- Server: DESKTOP-VOHEVGT\SQLFINALEXPRESS
- Authentication: Windows NT Integrated Security (SSPI)
- Initial Catalog: CarSalesDW
- Impersonation: Use the service account
- Data Source Name: Car Sales DW

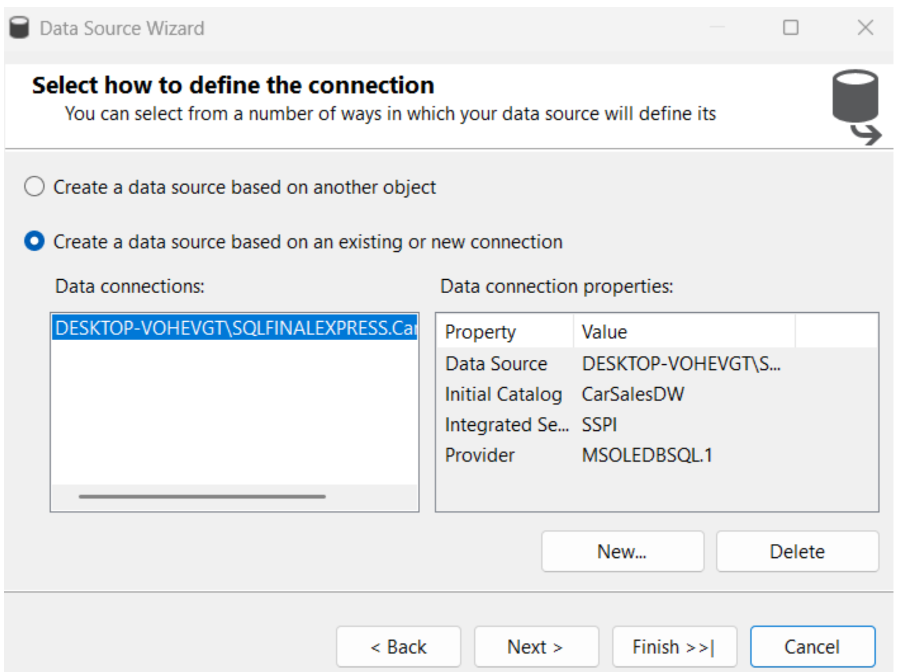


Figure 5 - Data Source Wizard - connection configuration to CarSalesDW

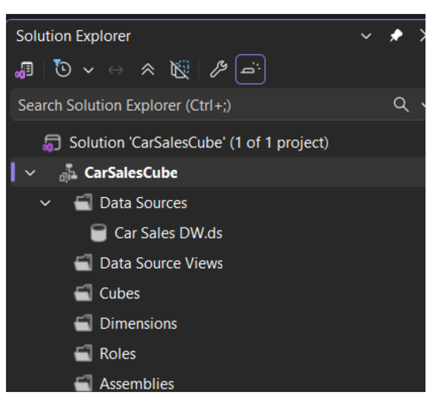


Figure 6 - Data Source created successfully

2.2 Create Data Source View (DSV)

A Data Source View was created by right-clicking 'Data Source Views' in Solution Explorer and selecting 'New Data Source View'. The following tables were selected from the CarSalesDW database:

- Fact_VehicleSales (Fact Table)
- Dim_Vehicle (Dimension Table)
- Dim_Seller (Dimension Table)
- Dim_Date (Dimension Table)
- Dim_Location (Dimension Table)

Note: Dim_Category was excluded as the category_key was removed from the Fact table due to unavailability of consistent body type data in the transactional dataset during ETL integration.

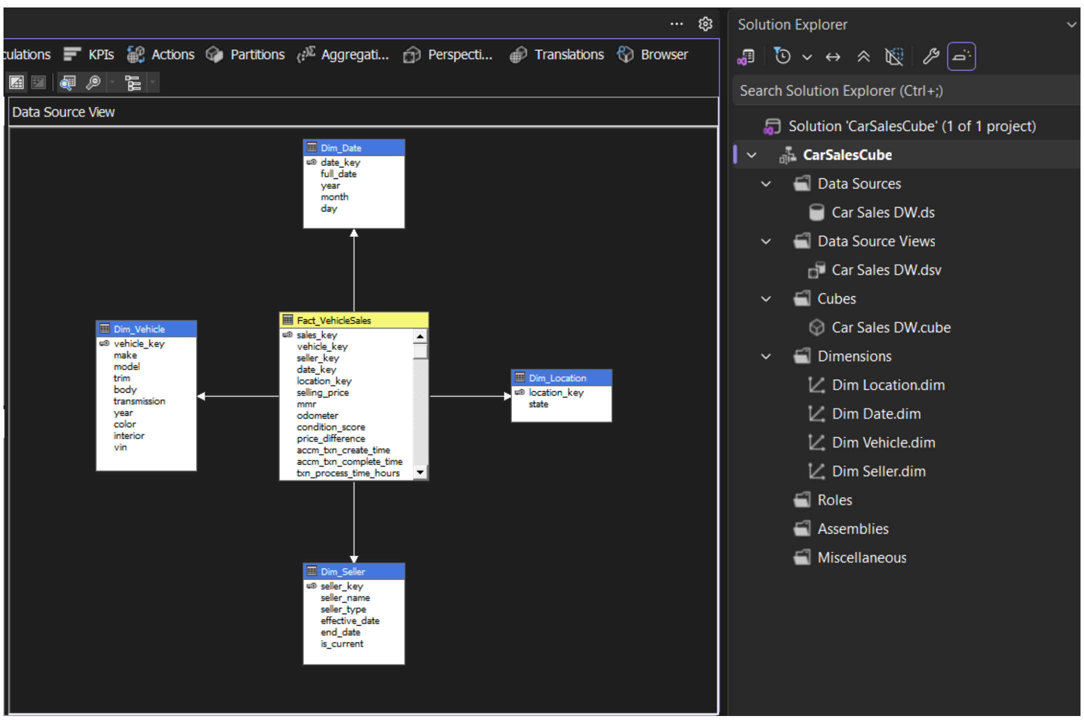


Figure 7 - Data Source View showing star schema – Fact_VehicleSales connected to all dimension tables

2.3 Create the Cube

A new Cube was created by right clicking the 'Cubes' folder in Solution Explorer and selecting 'New Cube'. The Cube Wizard was used with the following configuration:

- Creation Method: Use existing tables
- Measure Group Table: Fact_VehicleSales

The following measures were selected from the Fact_VehicleSales table:

Measure	Description
Selling Price	The actual selling price of the vehicle
mmr	Manheim Market Report reference price
Odometer	Vehicle odometer reading at time of sale
Condition Score	Vehicle condition score (1-50)
Price Difference	Difference between selling price and MMR
Txn Process Time Hours	Transaction processing time in hours
Fact Vehicle Sales Count	Count of vehicle sales transactions

The following dimensions were automatically detected and added:

- Dim_Location
- Dim_Date
- Dim_Vehicle
- Dim_Seller

Note: 'Fact Vehicle Sales' was unchecked from the dimensions list as it is the fact table, not a dimension.

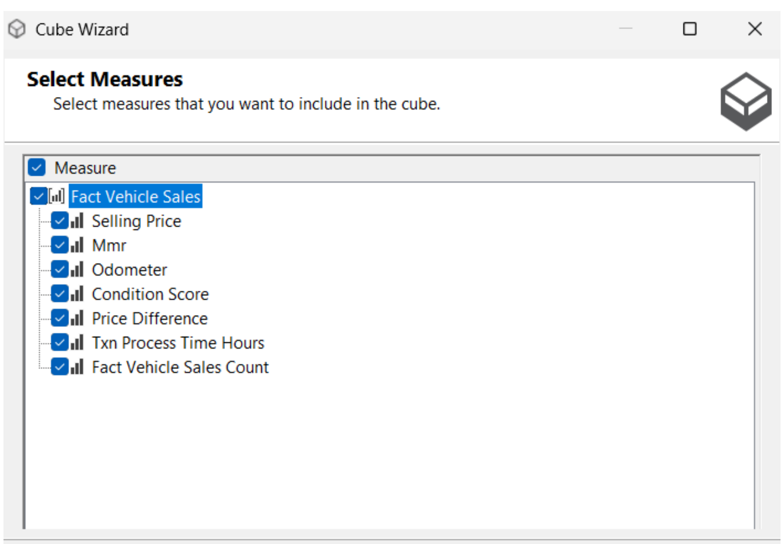


Figure 8 - Cube Wizard - select measures showing all measures selected

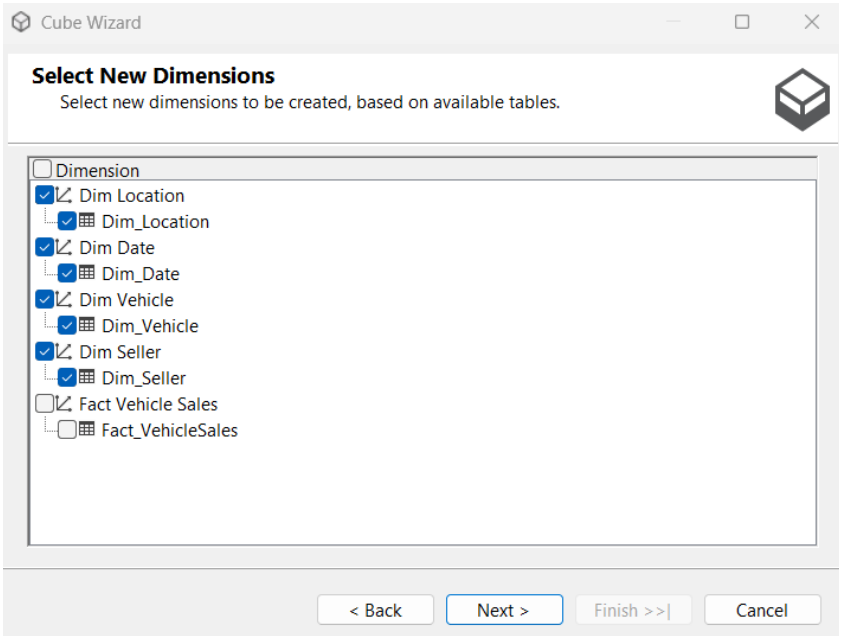


Figure 9 - Cube Wizard - Select Dimensions showing Dim Location, Dim Date, Dim Vehicle and Dim Seller

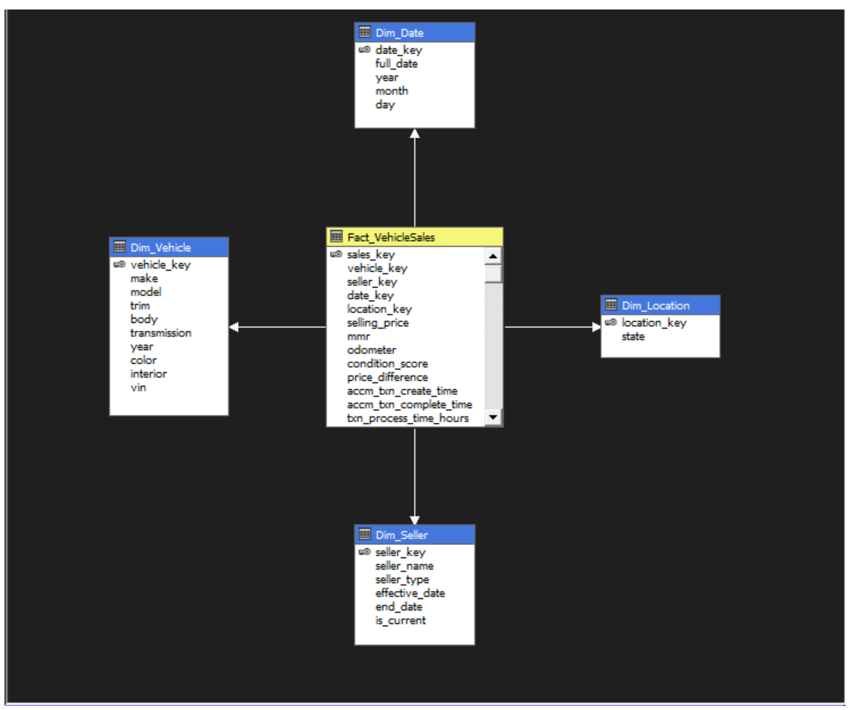


Figure 10 - Star Schema with Fact_VehicleSales and all dimension tables

2.4 Implementing Hierarchies

Hierarchies were implemented in dimension tables to support OLAP drill-down and roll-up operations. At least two hierarchies were created as described below.

Hierarchy 1: Year-Month-Day (Dim Date)

The Dim Date dimension was opened in the Dimension Designer. The following attributes were dragged from the Attributes pane to the Hierarchies pane in order:

- Level 1: Year
- Level 2: Month
- Level 3: Day

The hierarchy was renamed to 'Year-Month-Day'. This hierarchy enables time-based drill-down analysis from yearly summaries down to individual days.

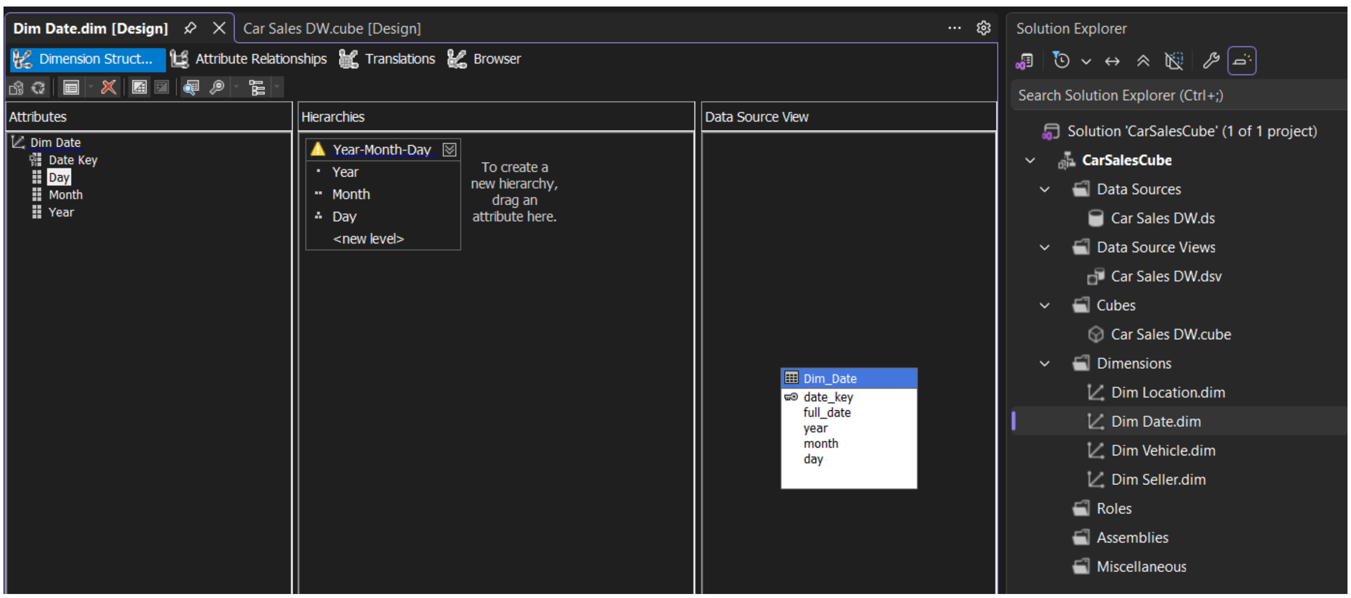


Figure 11 - Dim Date Hierarchy: Year-Month-Day created in Dimension Designer

Hierarchy 2: Make-Model-Trim (Dim Vehicle)

The Dim Vehicle dimension was opened, and the following attributes were added to the Attributes pane by dragging from the Data Source View, then arranged into a hierarchy:

- Level 1: Make (Vehicle manufacturer e.g., BMW, Audi)
- Level 2: Model (Vehicle model e.g., 3 Series)
- Level 3: Trim (Vehicle trim level e.g., Base, LX)

The hierarchy was renamed to 'Make-Model-Trim'. This hierarchy enables vehicle-based drill-down from manufacturer level to specific trim variants.

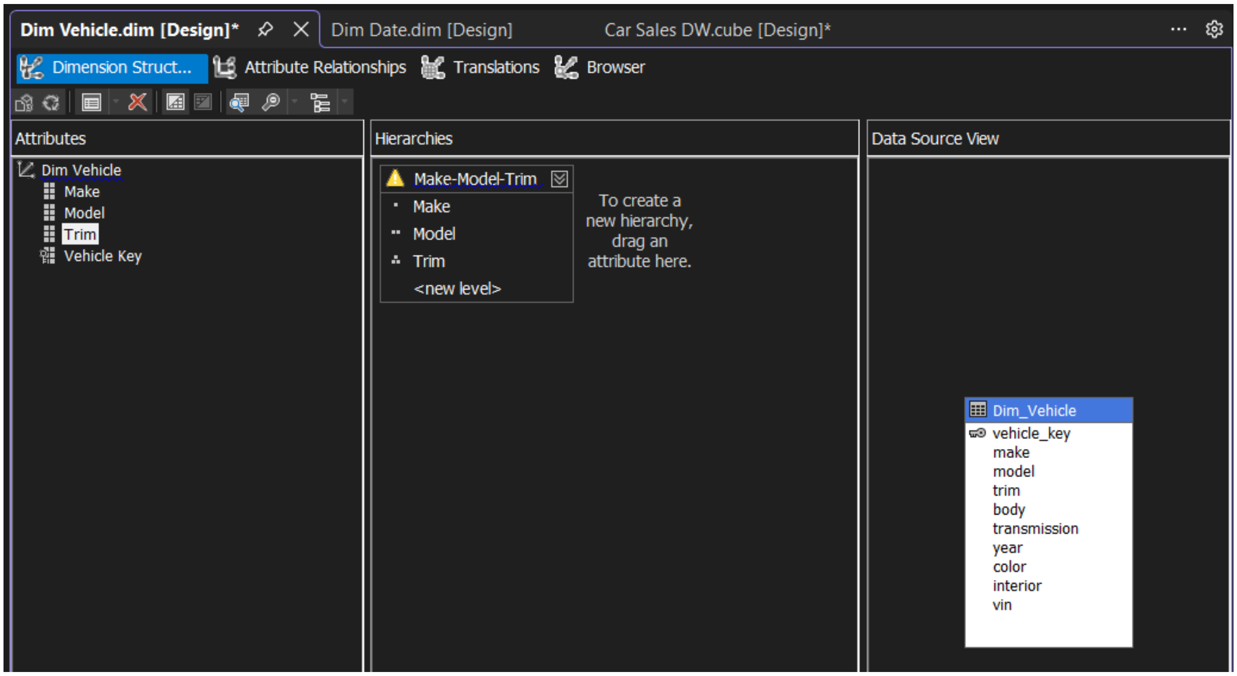


Figure 12 - Dim Vehicle Hierarchy: Make-Model-Trim created in Dimension Designer

2.5 Deploying the Cube

Before deploying, the deployment target was configured in the project Properties:

- Target Server: DESKTOP-VOHEVGT\SQLDEV
- Target Database: CarSalesCube

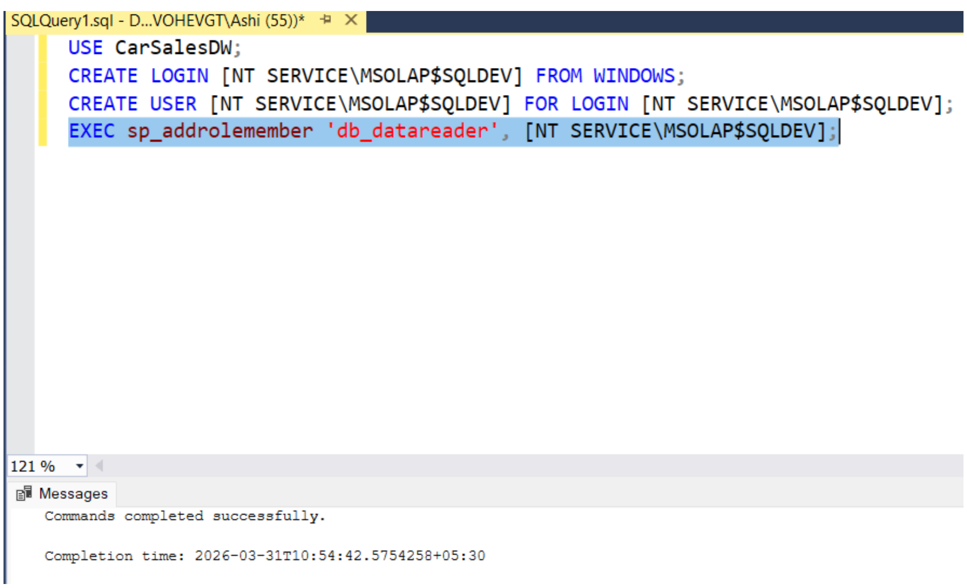


Figure 13 - SQL query executed on SQLFINALEXPRESS to grant the necessary permissions

After granting permissions, the project was deployed by right-clicking the CarSalesCube project in Solution Explorer and selecting 'Deploy'. The deployment completed successfully.

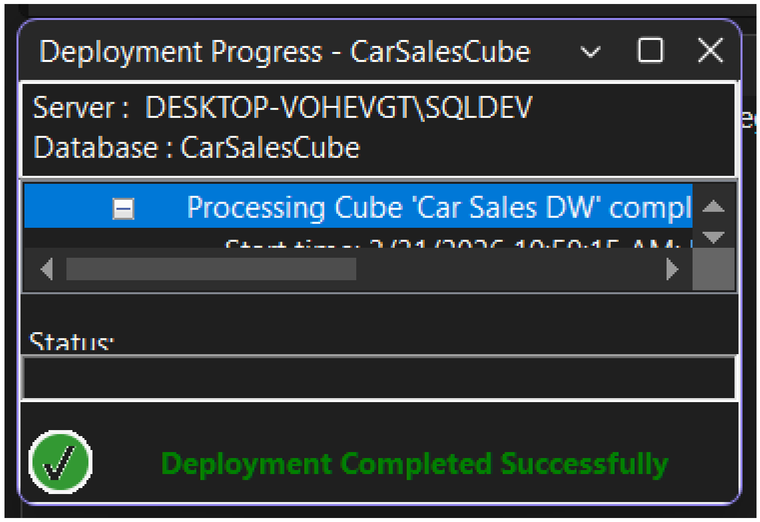


Figure 14 - Deployment Completed Successfully

3. Demonstration of OLAP Operations

3.1 Connecting Excel to the SSAS Cube

Microsoft Excel was used to connect to the deployed SSAS Cube and demonstrate OLAP operations using PivotTables and Pivot Charts.

The connection was established using Data tab -> Get Data -> From Database -> From Analysis Services

- Server: DESKTOP-VOHEVGT\SQLDEV
- Authentication: Use Windows Authentication
- Database: CarSalesCube
- Cube: Car Sales DW

A PivotTable Report was created on the existing worksheet to enable interactive OLAP analysis.

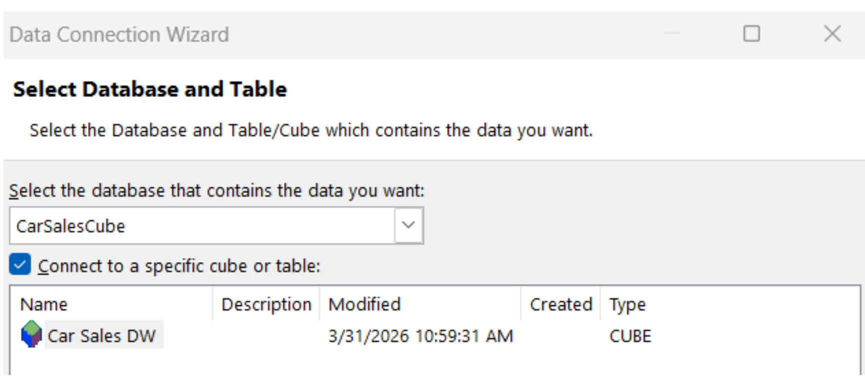


Figure 15 - Excel Data Connection Wizard - Connected to DESKTOP-VOHEVGT\SQLDEV, CarSalesCube

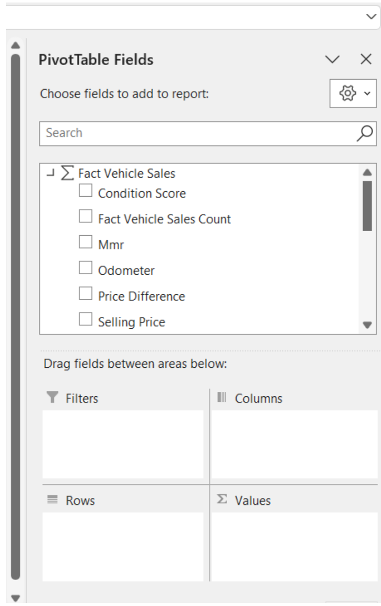


Figure 16 - Excel PivotTable Fields panel

3.2 OLAP Operation 1: Roll-Up

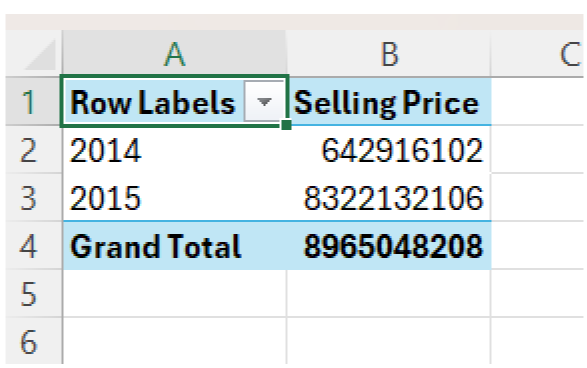
Definition

Roll-Up aggregates data by climbing up a concept hierarchy or by dimension reduction. It summarizes detailed data into higher-level summaries.

Demonstration

The Year field from Dim Date was placed in the Rows area, and Selling Price was placed in the Values area. This aggregated all vehicle sales data to the Year level, showing total selling prices for 2014 and 2015.

- Rows: Year (from Dim Date)
- Values: Selling Price (Sum)
- Result: 2014 = 642,916,102 | 2015 = 8,322,132,106 | Grand Total = 8,965,048,208



Row Labels	Selling Price
2014	642916102
2015	8322132106
Grand Total	**8965048208**

Figure 17 - Roll-Up: PivotTable showing Selling Price aggregated by Year (2014, 2015, Grand Total)

3.3 OLAP Operation 2: Drill-Down

Demonstration

The Month field was added below Year in the Rows area, drilling down from Year level to Month level within each year, revealing monthly sales breakdowns.

- Rows: Year > Month (using Year-Month-Day hierarchy)
- Values: Selling Price (Sum)
- Result: Year 2014 expanded to show months 1, 2, 12; Year 2015 expanded to show months 1-8

	A	B
1	Row Labels	Selling Price
2	⊟ 2014	
3	1	3923995
4	12	638866035
5	2	126072
6	⊟ 2015	
7	1	2096580055
8	2	2591763287
9	3	925231877
10	4	27681027
11	5	792761913
12	6	1781999184
13	7	105656849
14	8	457914
15	Grand Total	8965048208

Figure 18 - Drill-Down: PivotTable showing Selling Price broken down by Year then Month

3.4 OLAP Operation 3: Slice

Definition

Slice selects one specific value for one dimension, creating a subset of the cube. It reduces the data to a single dimension value, effectively slicing through the cube.

Demonstration

The Make field from Dim Vehicle was placed in the Filters area and filtered to 'Audi', showing only Audi vehicle sales across all time periods.

- Filter (Slice): Make = Audi
- Rows: Year > Month

- Values: Selling Price (Sum)
- Result: Only Audi sales visible - Grand Total = 137,517,399

Make	Audi
Row Labels	**Selling Price**
2014	
12	10274749
2015	
1	33204477
2	35532361
3	8913477
4	217049
5	14714275
6	32983717
7	1677294
Grand Total	**137517399**

Figure 19 - PivotTable filtered by Make = Audi showing only Audi vehicle sales

3.5 OLAP Operation 4: Dice

Definition

Dice selects two or more dimensions and filters values for each, creating a sub-cube from the original cube. It applies multiple dimension filters simultaneously.

Demonstration

Two filters were applied simultaneously - Make set to 'BMW' and Model set to '6 Series Gran Coupe', creating a sub-cube showing only that specific BMW model's sales data.

- Filter 1 (Dice): Make = BMW
- Filter 2 (Dice): Model = 6 Series Gran Coupe
- Rows: Year > Month
- Values: Selling Price (Sum)
- Result: BMW 6 Series Gran Coupe sales only - Grand Total = 7,008,959

Make	BMW
Model	6 Series Gran Coupe
Row Labels	**Selling Price**
2014	
12	955502
2015	
1	667268
2	1339114
3	175781
5	1443530
6	2263457
7	164307
Grand Total	**7008959**

Figure 20 - Dice: PivotTable filtered by Make = BMW AND Model = 6 Series Gran Coupe

3.6 OLAP Operation 5: Pivot

Definition

Pivot rotates the data cube to provide an alternative presentation of the data. It reorients the multidimensional view by swapping rows and columns to gain a different perspective.

Demonstration

The axes were rotated by moving Year to the Columns area and placing Make-Model-Trim hierarchy and Month in the Rows area. This provided a different view with years as column headers and vehicle/time details as rows.

- Columns: Year (2014, 2015)
- Rows: Month, Make-Model-Trim
- Filters: Make = BMW, Model = 6 Series Gran Coupe
- Values: Selling Price (Sum)
- Result: Years shown as columns, vehicle and time details as rows

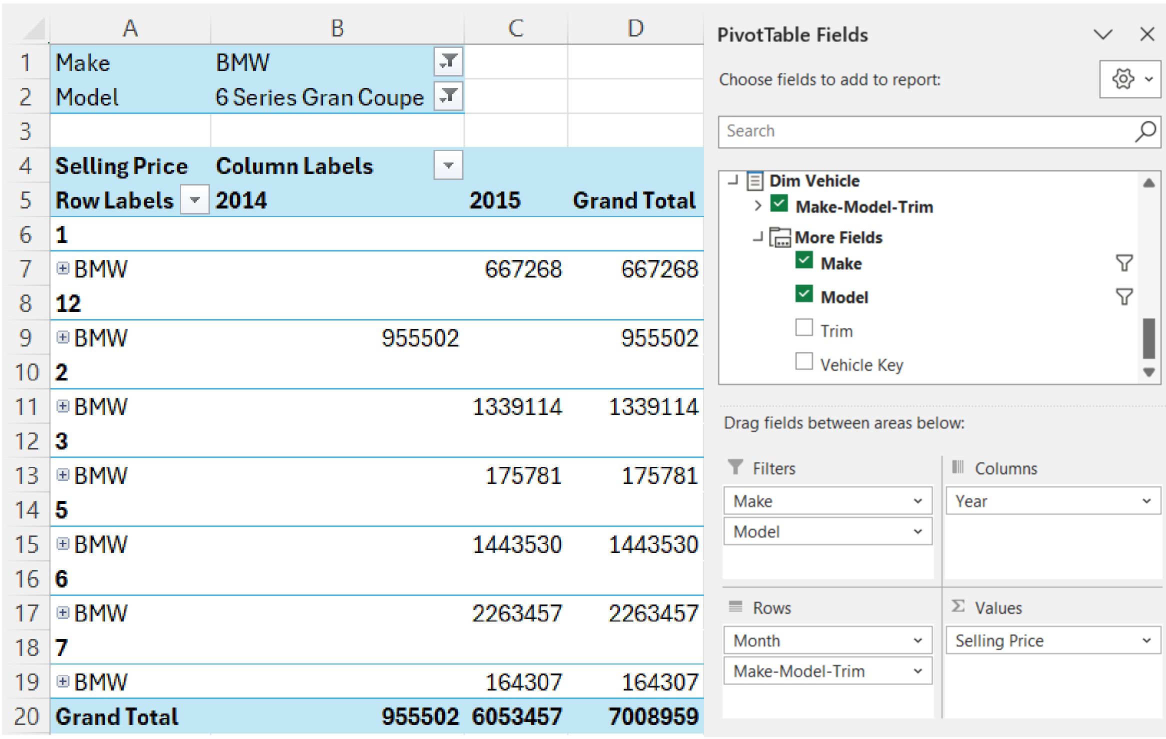


Figure 21 - Pivot: PivotTable with Year as Columns and Make-Model-Trim + Month as Rows - rotated view

3.7 Pivot Chart Visualization

A PivotChart was created from the PivotTable to provide a visual representation of the OLAP data. The chart was created using PivotTable Analyze tab > PivotChart.

- Chart Type: Clustered Bar Chart
- X-axis: Month with Make (BMW)
- Y-axis: Selling Price
- Legend: Year (2014 = Blue, 2015 = Orange)
- Filters: Make = BMW, Model = 6 Series Gran Coupe

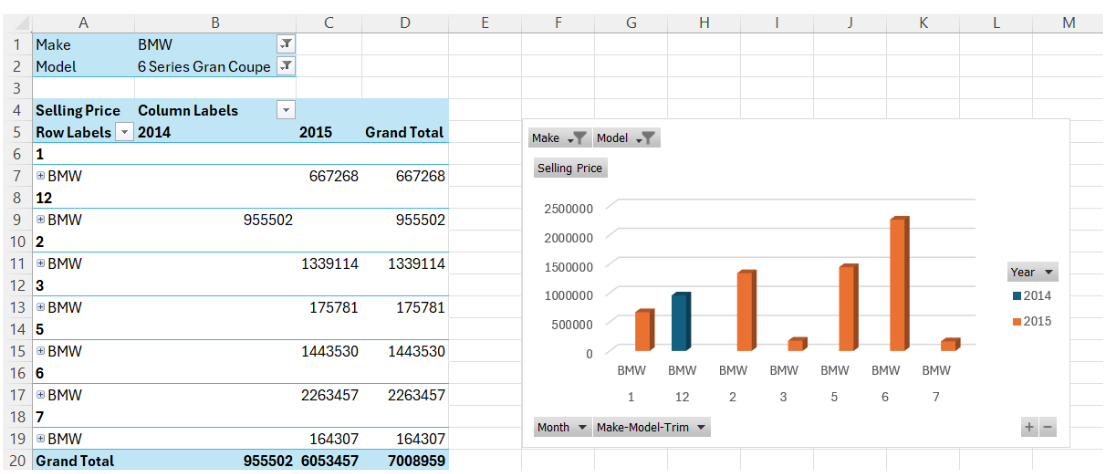


Figure 22 - PivotChart: Bar chart showing BMW 6 Series Gran Coupe Selling Price by Month, colored by Year

4. Power BI Report - Car Sales Data Warehouse

This report presents the development of interactive dashboards using Power BI based on the CarSalesDW data warehouse. The objective is to transform raw data into meaningful insights using visual analytics, enabling users to explore vehicle sales performance across multiple dimensions such as time, location, vehicle attributes, and sellers.

The reports were designed to support decision-making by providing both high-level summaries and detailed drill-down capabilities.

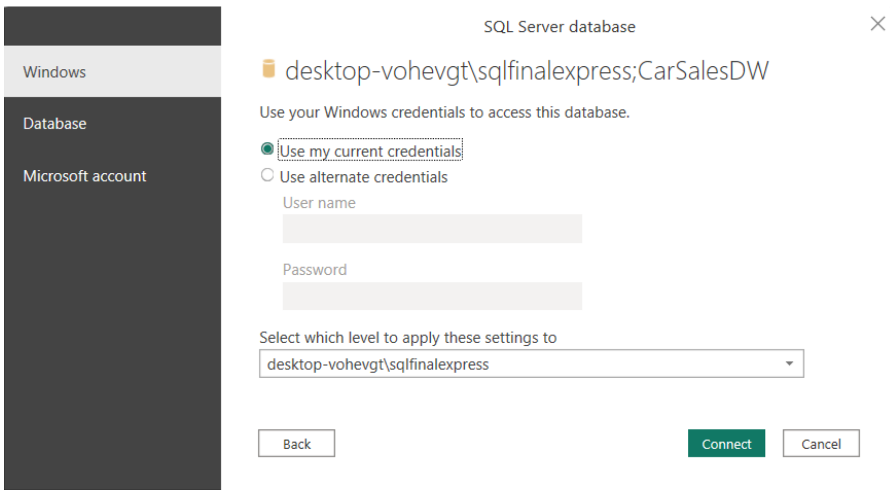


Figure 23 - Connecting SQLFINALEXPRESS to Power BI

4.1 Data Preparation

The data used for visualization was sourced from the CarSalesDW data warehouse, which was previously developed using ETL processes.

The dataset includes:

- Fact table: **Fact_VehicleSales**
- Dimension tables:
    - ❖ **Dim_Date**
    - ❖ **Dim_Vehicle**
    - ❖ **Dim_Seller**
    - ❖ **Dim_Location**

Relationships were established using surrogate keys such as date_key, vehicle_key, seller_key, and location_key. The data model follows a **star schema**, ensuring efficient querying and aggregation.

Data cleaning steps included:

- Removing irrelevant values such as “unknown”

- Formatting date fields
- Ensuring correct data types for numerical and categorical attributes

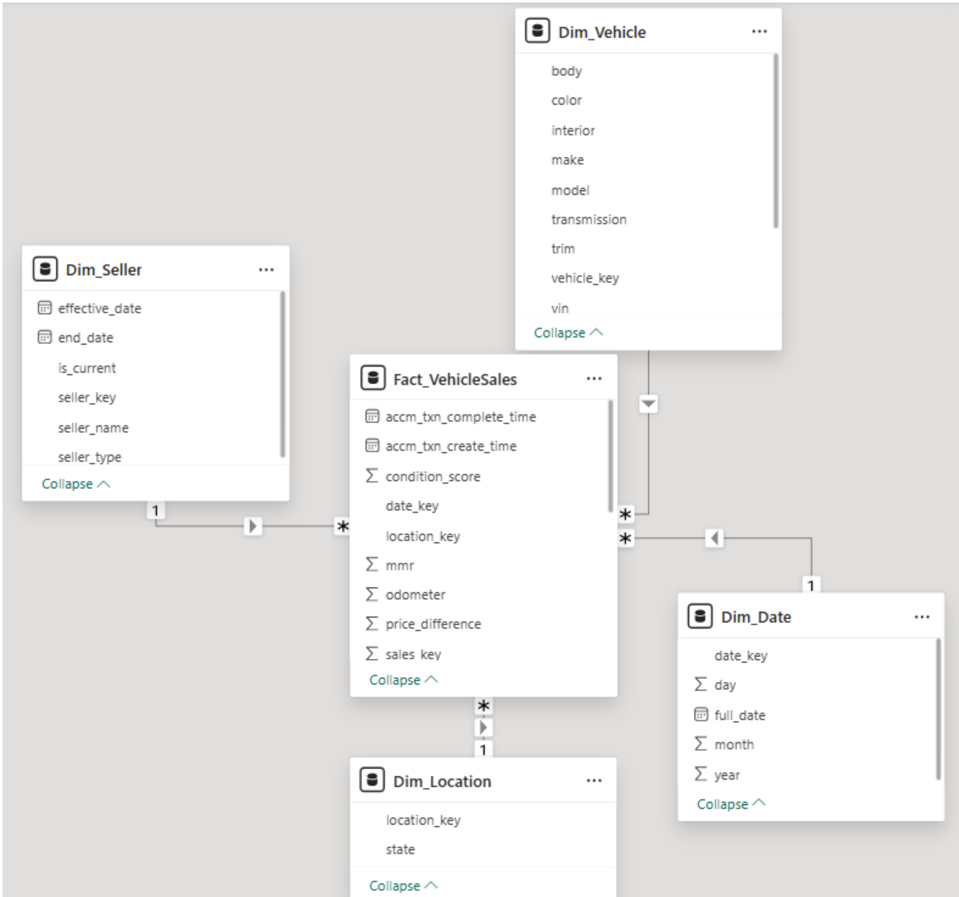


Figure 24 - Star Schema Model View in Power BI

4.2 Data Modeling

The data model was implemented as a star schema, consisting of one central fact table and multiple dimension tables.

- **Fact_VehicleSales** contains transactional data such as selling price, condition score, odometer, and transaction timestamps.
- **Dim_Date** enables time-based analysis (year, month, day).
- **Dim_Vehicle** provides vehicle attributes such as make, model, transmission, and color.
- **Dim_Seller** stores seller-related details.
- **Dim_Location** supports geographical analysis.

This structure allows efficient aggregation and supports advanced analytics such as drill-down and drill-through.

4.3 DAX Measures

Several DAX measures were created to support analysis:

1. To calculate Total sales

```
Total Sales = COUNT(Fact_VehicleSales[txn_id])
```

This measure calculates the total number of vehicle sales transactions.

2. To calculate Total Revenue

```
Total Revenue = SUM(Fact_VehicleSales[selling_price])
```

This measure calculates the total revenue generated from vehicle sales

3. To calculate total models

```
Total Models = DISTINCTCOUNT(Dim_Vehicle[model])
```

This measure counts the number of unique vehicle models available in the dataset.

4. To calculate Minimum Price

```
Min Price = MIN(Fact_VehicleSales[selling_price])
```

This measure identifies the lowest selling price of vehicles.

5. To calculate Maximum Price

```
Max Price = MAX(Fact_VehicleSales[selling_price])
```

This measure identifies the highest selling price of vehicles.

6. To calculate Average price

```
Average Price = AVERAGE(Fact_VehicleSales[selling_price])
```

This measure calculates the average selling price of vehicles.

7. To calculate total brands

```
Total Brands = DISTINCTCOUNT(Dim_Vehicle[make])
```

This measure counts the number of distinct vehicle brands.

8. To create Dynamic title for Drill-through page

```
Dynamic Title =
"Vehicle Details - " &
SELECTEDVALUE(Dim_Vehicle[model], "All Models")
```

This measure dynamically updates the title of the drill-through page based on the selected vehicle model.

These measures enabled aggregation, comparison, and percentage-based analysis across different dimensions.

4.4 Report 1 – Matrix Report

A matrix visual was created to display detailed tabular data with row and column groupings.

- Rows: Vehicle makes
- Columns: Year
- Values: Total sales

This report allows users to compare sales performance across different brands and years, providing a structured overview of the data.

make	2014	2015	Total
dot	500.00		**500.00**
dodge tk	550.00		**550.00**
Daewoo	300.00	900.00	**1,200.00**
chev truck	2,000.00		**2,000.00**
hyundai tk	2,100.00		**2,100.00**
ford truck	350.00	2,084.00	**2,434.00**
ford tk	2,800.00		**2,800.00**
Geo	300.00	12,250.00	**12,550.00**
mazda tk	6,500.00	6,212.00	**12,712.00**
mercedes-b	18,500.00	8,800.00	**27,300.00**
airstream	71,000.00		**71,000.00**
gmc truck	38,659.00	53,784.00	**92,443.00**
Lotus		163,242.00	**163,242.00**
plymouth	6,700.00	264,480.00	**271,180.00**
Isuzu	37,400.00	361,201.00	**398,601.00**
vw	6,050.00	409,687.00	**415,737.00**
Fisker		418,150.00	**418,150.00**
Oldsmobile	76,832.00	353,450.00	**430,282.00**
landrover	233,574.00	206,786.00	**440,360.00**
Lamborghini		450,500.00	**450,500.00**
mercedes	369,638.00	282,315.00	**651,953.00**
Aston Martin		1,567,402.00	**1,567,402.00**
Tesla	160,053.00	1,541,620.00	**1,701,673.00**
Saab	243,092.00	1,827,988.00	**2,071,080.00**
smart	261,872.00	2,653,271.00	**2,915,143.00**
Ferrari	434,000.00	2,646,766.00	**3,080,766.00**
Rolls-Royce	149,800.00	3,075,034.00	**3,224,834.00**
Suzuki	573,830.00	4,378,249.00	**4,952,079.00**
Maserati	376,600.00	6,935,942.00	**7,312,542.00**
Mercury	1,020,229.00	8,250,835.00	**9,271,064.00**
Bentley	1,317,709.00	8,261,978.00	**9,579,687.00**
FIAT	457,525.00	10,278,337.00	**10,735,862.00**
Saturn	1,465,035.00	10,026,264.00	**11,491,299.00**
Total	**642,916,102.00**	**8,322,132,106.00**	**8,965,048,208.00**

Figure 25 - Matrix Report

4.5 Report 2 – Slicers and Interactive Dashboard

An interactive dashboard was developed using multiple slicers and visualizations.

Slicers:

- Year
- Make
- Model

These slicers act as cascading filters, where selecting a value dynamically filters other visuals.

Visualizations used:

- KPI Cards (Total Revenue, Total Sales, Average Price, Total Models)
- Bar Chart (Top 10 Car Brands)
- Donut Chart (Sales by Transmission)
- Tree map (Sales by Model)
- Matrix Table (Detailed data view)

This report provides a comprehensive overview of vehicle sales and enables users to interactively explore the data.

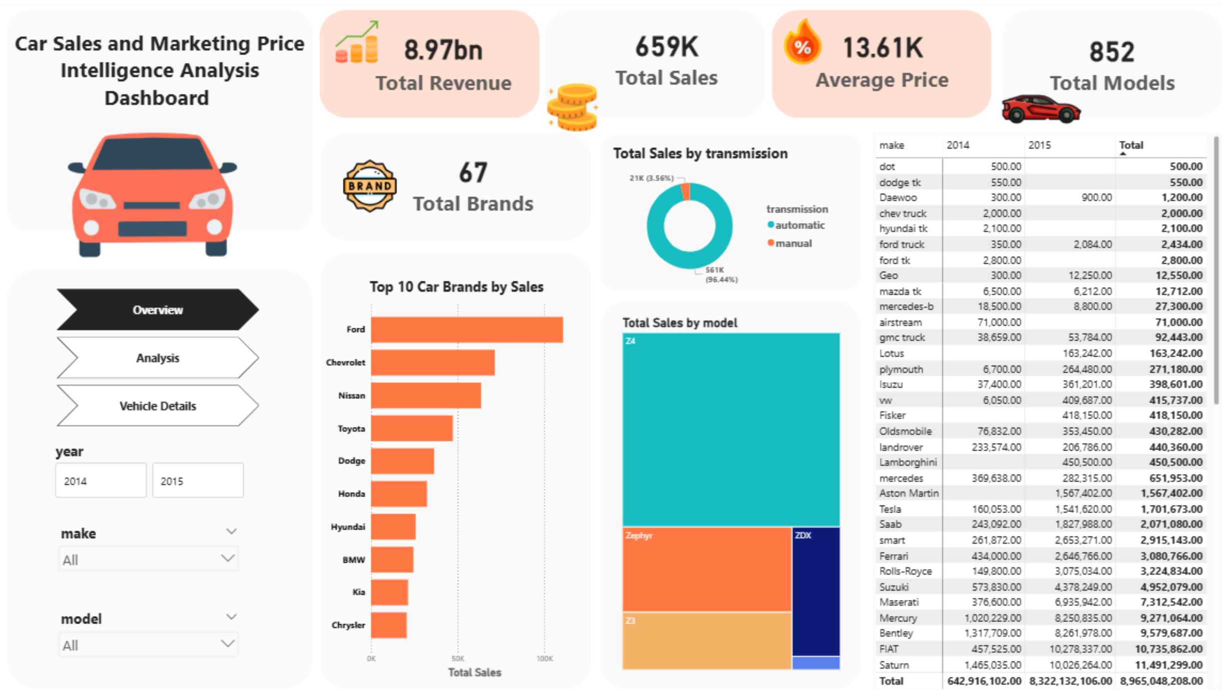


Figure 26 - Power BI Overview Report Page

4.6 Report 3 - Drill-Down Report

A drill-down report was implemented using a column/line chart with a date hierarchy.

Hierarchy levels:

- Year → Month → Day

Users can interactively drill down from yearly summaries to monthly and daily data by clicking on the visual.

This feature allows deeper analysis of sales trends over time.

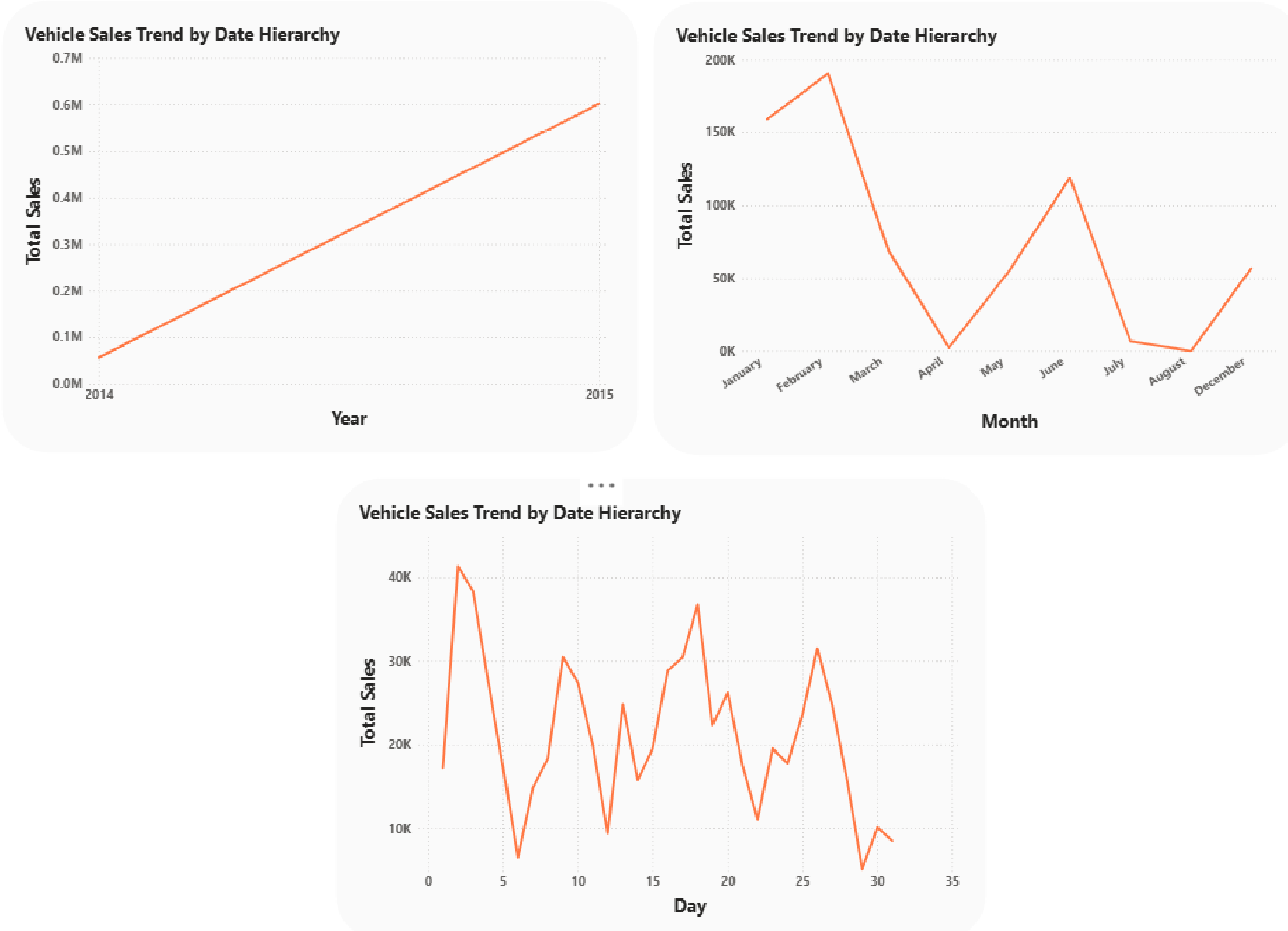


Figure 27 - Vehicle sales trend by date hierarchy drill-down report

4.7 Report 4 – Drill-Through Report

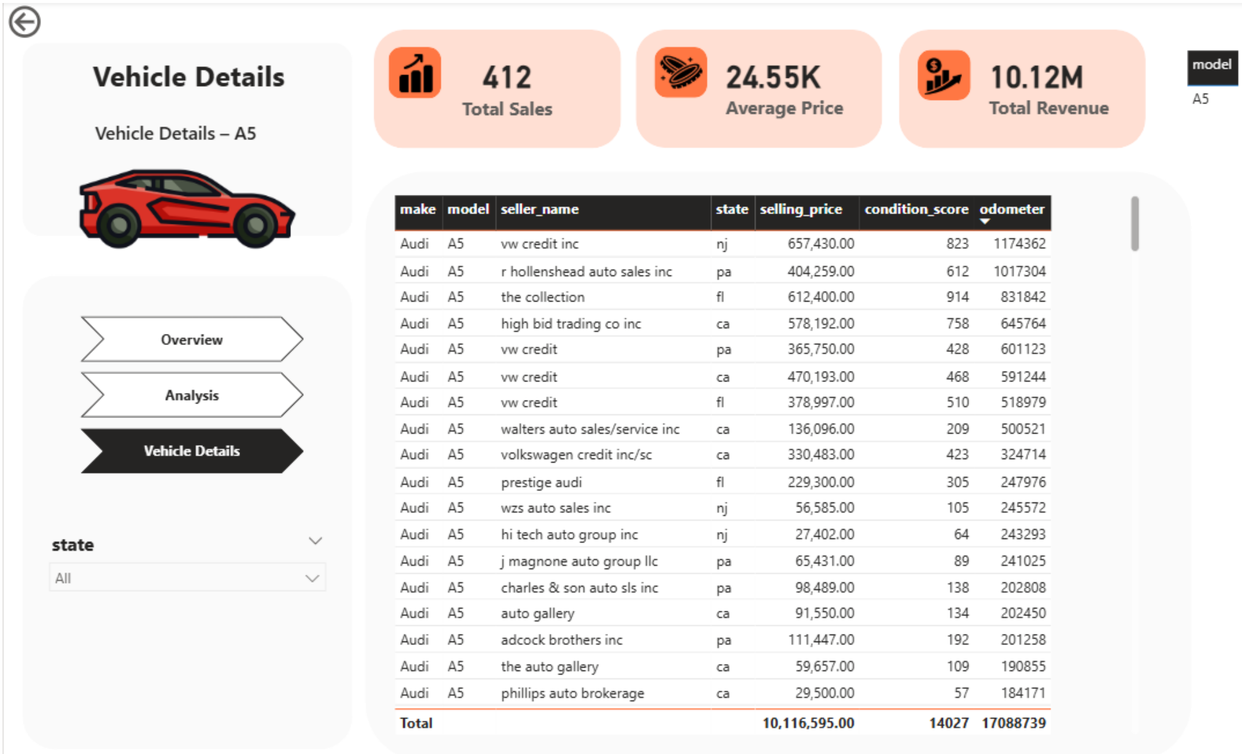
A drill-through report was created to allow users to navigate from summary visuals to detailed information.

- Drill-through field: Vehicle model
- Users can right-click on a model and navigate to a detailed page

Drill-through page includes:

- KPI cards (Total Sales, Average Price, Total Revenue)
- Detailed table (make, model, seller, state, price, condition, odometer)

The selected model is passed as a filter, ensuring that only relevant data is displayed.



make	model	seller_name	state	selling_price	condition_score	odometer
Audi	A5	vw credit inc	nj	657,430.00	823	1174362
Audi	A5	r hollenshead auto sales inc	pa	404,259.00	612	1017304
Audi	A5	the collection	fl	612,400.00	914	831842
Audi	A5	high bid trading co inc	ca	578,192.00	758	645764
Audi	A5	vw credit	pa	365,750.00	428	601123
Audi	A5	vw credit	ca	470,193.00	468	591244
Audi	A5	vw credit	fl	378,997.00	510	518979
Audi	A5	walters auto sales/service inc	ca	136,096.00	209	500521
Audi	A5	volkswagen credit inc/sc	ca	330,483.00	423	324714
Audi	A5	prestige audi	fl	229,300.00	305	247976
Audi	A5	wzs auto sales inc	nj	56,585.00	105	245572
Audi	A5	hi tech auto group inc	nj	27,402.00	64	243293
Audi	A5	j magnone auto group llc	pa	65,431.00	89	241025
Audi	A5	charles & son auto sls inc	pa	98,489.00	138	202808
Audi	A5	auto gallery	ca	91,550.00	134	202450
Audi	A5	adcock brothers inc	pa	111,447.00	192	201258
Audi	A5	the auto gallery	ca	59,657.00	109	190855
Audi	A5	phillips auto brokerage	ca	29,500.00	57	184171
Total				10,116,595.00	14027	17088739

Figure 28 - Drill-through report

4.8 Visual Design

A consistent and modern design theme was applied across all dashboards:

- Light background with soft color palette
- Highlight colors used for key metrics
- Consistent font styles and spacing
- Use of icons and card visuals for clarity

The layout was designed to enhance readability and user experience.

4.9 Interactivity Features

The dashboards include several interactive features:

- Slicers for dynamic filtering
- Drill-down for hierarchical exploration
- Drill-through for detailed analysis
- Page navigation for easy movement between reports

These features allow users to explore the data intuitively and gain insights efficiently.

Figure 29 - Power BI Analysis report page

The Power BI reports successfully transform raw data into meaningful insights through interactive visualizations. The use of DAX, star schema modeling, and advanced features such as drill-down and drill-through enhances analytical capabilities.

The dashboards provide a comprehensive view of vehicle sales performance and support data-driven decision-making.

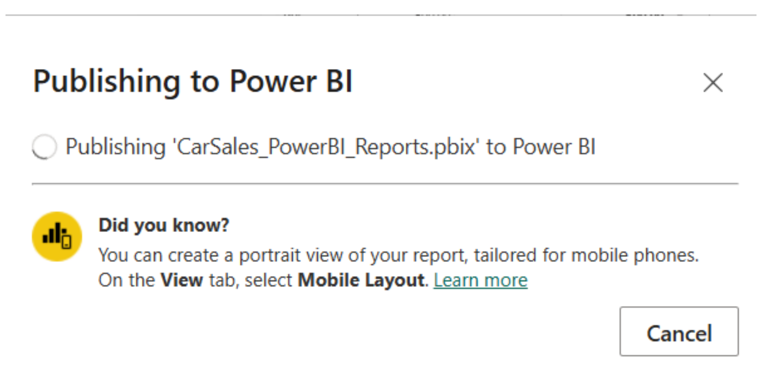


Figure 30 - Publishing the power BI report

The Power BI reports were successfully published to the Power BI Service, allowing online access and demonstration of interactive dashboards.

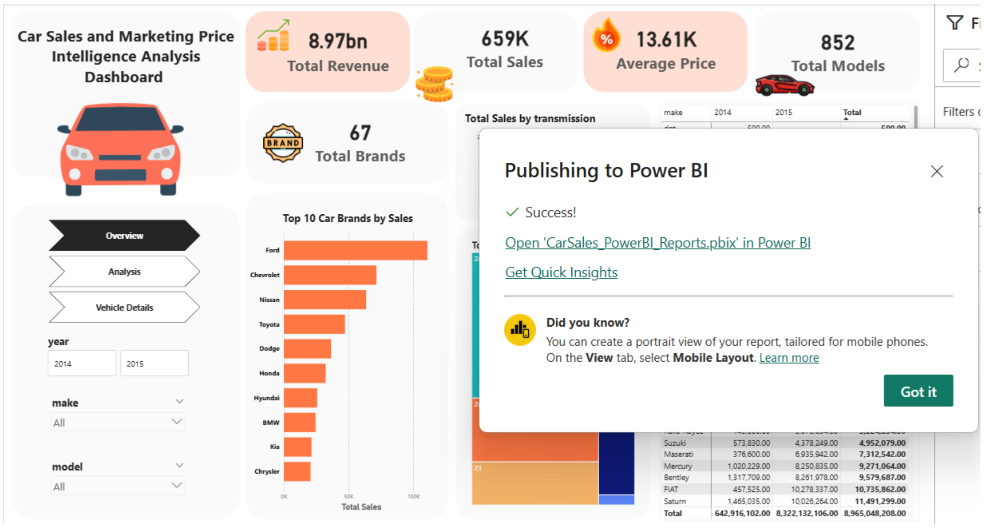


Figure 31 - Power BI report publish successfully